Core Modules - Detailed Explanation (Paragraph Style)

1. Authentication Module

The Authentication Module serves as the gateway to the entire Job Tracking System, handling user registration, login, and session management across all three user roles. When a new user visits the platform, they begin by creating an account where they select their role as either a Job Seeker, Recruiter, or Admin. The system captures essential credentials including username, password, and role type, then validates this information on the server side to prevent duplicate usernames and enforce password security standards. Once registered, users can log in with their credentials, and upon successful validation, the system creates a session that persists using localStorage, allowing users to remain authenticated across page refreshes and browser sessions. The module implements role-based access control, meaning each user type is directed to a specialized dashboard with permissions tailored to their needs. Job seekers see application tracking tools, recruiters access candidate management features, and admins get platform oversight capabilities. This module ensures security through credential validation, session persistence, and proper route protection on both frontend and backend.

2. Job Tracking Module

The Job Tracking Module forms the core functionality for job seekers, providing a comprehensive personal application management system where users can monitor every opportunity they pursue. When a user finds a job they're interested in, they can add it to their tracker by entering details such as company name, role title, application status, deadline date, follow-up reminders, personal notes, and the job posting URL. The system supports full CRUD operations, allowing users to create new entries, view their complete application history, update status as they progress through interviews, and delete entries for opportunities they no longer wish to track. The status field follows a natural workflow from Applied through Interview, Offer, Rejected, or Withdrawn states. The module includes deadline management features that help users stay on top of upcoming dates, and all data is isolated per user through MongoDB queries filtered by user_id, ensuring privacy and data separation. This module essentially replaces the scattered spreadsheets and sticky notes that job seekers traditionally use, centralizing everything in one accessible dashboard.

3. Resume Builder Module

The Resume Builder Module offers a structured approach to creating professional resumes through six comprehensive sections that capture all essential career information. Users begin by filling in their personal information including full name, contact email, phone number, and location. They then craft a professional summary that serves as their elevator pitch to potential employers. The work experience section allows for multiple entries where users detail their previous roles, companies, employment dates, and key achievements using action-oriented descriptions. The education section captures academic background with institution names, degrees earned, graduation years, and GPA if desired. Users can build their skills profile through tag-based input featuring over forty predefined technical and soft skills, or add custom skills relevant to their field. Finally, the projects section showcases portfolio work with project titles, detailed descriptions, technologies used, and links to live demos or repositories. The module supports creating multiple resume versions tailored to different job types, and all data persists in the MongoDB resumes collection with a nested schema that maintains relationships between sections while allowing flexible data structures for varying career histories.

4. Job Portal Module

The Job Portal Module creates a direct marketplace connecting job seekers with recruiters, functioning as the transactional heart of the platform. For recruiters, the module provides tools to create detailed job postings that include position titles, company information, comprehensive requirement lists, salary ranges, and location details. Once posted, recruiters can view all applications submitted for their positions, access attached documents, and move candidates through the hiring pipeline by updating application statuses. Job seekers benefit from a browsing interface where they can explore available positions, filter by relevant criteria, and submit applications with their selected resume and a tailored cover letter. The application process captures the job posting ID, applicant ID, selected resume, cover letter content, and initial status, storing everything in the job_applications collection for tracking. The module enables real-time status visibility, meaning applicants can see when a recruiter views their application or changes their status to accepted or rejected. This two-sided marketplace approach streamlines the traditional job application process by eliminating email chains and providing transparent status tracking for both parties.

5. Admin Module

The Admin Module provides platform oversight capabilities, giving administrators the tools needed to manage users, moderate content, and monitor system health. Admins can view comprehensive lists of all registered users across all roles, seeing usernames, registration dates, and activity levels. When necessary, they have the authority to delete user accounts, which triggers a cascade deletion of all associated data including job applications, resumes, and job postings to maintain data integrity. The module also includes content moderation features allowing admins to review and remove inappropriate job postings that violate platform guidelines. System statistics give admins visibility into platform growth through metrics like total registered users, active job postings, applications submitted, and overall engagement rates. This administrative oversight ensures the platform maintains quality standards, prevents abuse, and provides insights for scaling decisions. Access to this module is strictly restricted to users with admin role designation, protecting sensitive user management functions from unauthorized access.

6. File Upload Module

The File Upload Module handles document management for the job application process, specifically managing resume and cover letter attachments. The system uses Base64 encoding to convert binary file data into text strings that can be stored directly in MongoDB documents. When a user selects a file for upload, the frontend JavaScript reads the file, converts it to a Base64 string, and sends this data along with the filename to the backend API. The server stores this information in the job_seeker_resumes collection, capturing the user ID, document name, file type, and the Base64 content string. The module enforces a ten megabyte size limit to prevent database bloat and ensure reasonable storage usage. When recruiters review applications, they can retrieve these documents by requesting the Base64 data from the API, which the frontend then decodes and presents as downloadable files or inline previews depending on the file type. While Base64 storage works well for the prototype phase, the architecture is designed to easily transition to cloud storage solutions like AWS S3 or Google Cloud Storage as the platform scales.

7. Skill Analysis Module

The Skill Analysis Module adds intelligent processing capabilities to the platform by automatically extracting and analyzing skills from job descriptions and resumes. The system maintains a database of over forty predefined technical skills ranging from programming languages like JavaScript, Python, and Java to frameworks like React, Node.js, and Django, along with soft skills and methodologies. When a recruiter posts a job, the module parses the description text using keyword matching to identify

required skills, including contextual analysis that maps related terms to standardized skill names. For example, mentions of remote work get mapped to Remote Collaboration skills, while agile methodology references map to Agile/Scrum skills. The module can then compare the extracted job requirements against a job seeker's resume skills to identify matches and gaps. This gap analysis helps seekers understand what skills they might need to develop for specific roles, while recruiters benefit from better candidate filtering based on skill match percentages. The keyword-based approach provides immediate value and creates a foundation for future machine learning enhancements that could improve matching accuracy through natural language processing.

Summary

These seven modules work together to create a comprehensive job tracking ecosystem. The Authentication Module secures access and manages user roles, while the Job Tracking Module gives seekers personal application management tools. The Resume Builder helps users present their qualifications professionally, and the Job Portal connects these prepared candidates with recruiters who can post positions and manage applications. The Admin Module maintains platform integrity, the File Upload Module handles document exchanges, and the Skill Analysis Module brings intelligent matching capabilities. Together, they transform the chaotic job search process into a structured, trackable, and efficient experience for all user types.